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Produced Water Based Green Hydrogen Production Using an Anion Exchange Membrane Water Electrolyzer (AEMWE) System

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Abstract

Considerable volumes of produced water are generated along with hydrocarbons in oil & gas fields worldwide. Such vast quantities of produced water can be transformed from a waste stream that requires handling and disposal to a valuable resource that generates significant economic and environmental benefits. If this produced water can be recycled and utilized for low-carbon energy applications, it becomes a game changer to promote circular water economy, sustainability, and support the energy transition. Green hydrogen, the optimal energy carrier, plays a crucial role in connecting smart grids and scaling up low-carbon energy generation.

This study experimentally investigates the feasibility of using treated produced water obtained from zero liquid discharge (ZLD) desalination technology for H_2 generation using proton exchange membrane water electrolyzer (PEMWE) and anion exchange membrane water electrolyzer (AEMWE) systems. For evaluating the quality of treated produced water, the current density was measured by I-V measurement at controlled temperature (60°C). Also, it was measured with and without additional purification process to evaluate the suitability of treated produced water as direct resource for H_2 production.

The results from PEMWE showed different measured voltages with (1.8 V at 1.5 A/cm^2) and without (2.5 V at 1.5 A/cm^2) additional purification process. This finding indicates that impurities in treated produced water will have a negative effect on the performance of the PEMWE to lower the energy efficiency. On the other hand, the measured voltage from AEMWE showed consistent performance with (2.3 V at 1.5 A/cm^2) and without additional purification process. Such result confirms the robustness of AEMWE to generate H_2 with almost similar energy efficiency using treated produced water with and without further purification, when compared to PEMWE. This means the treated produced water of ZLD technology can be directly used with AEMWE without any additional purification process to sustain the operation costs and H_2 production efficiency. The results obtained from durability test and optimized electrode materials were also presented to demonstrate the long-term reliability of AEMWE for enhancing H_2 production efficiency with treated produced water.

This first-ever laboratory study demonstrates the successful and direct utilization of treated produced water using the AEMWE for green H_2 production. The upscaling of AEMWE technology has huge potential

to recycle produced water and it can also become a win-win technological solution to promote circular water/carbon economies and support the ongoing energy transition initiatives in the industry.

Introduction

The world is transitioning towards a low-carbon economy, driven by growing concerns over climate change, environmental sustainability, and energy security (Kabeyi and Olanrewaju 2022, Alshammari et al 2024). Hydrogen, as a clean-burning fuel, has garnered significant attention for its potential to decarbonize hard-to-abate sectors such as heavy industries, transportation, and power generation (Dawood et al 2020). Green hydrogen, produced from renewable energy sources through water electrolysis, represents a pivotal step towards realizing a carbon-neutral future (Holladay et al 2009, Terlouw et al 2022). However, the large volumes of freshwater required for conventional hydrogen production pose significant challenges, particularly in water-scarce regions where competition for availing this resource is already acute (Olaitan et al 2024).

The utilization of produced water, a byproduct of oil and gas operations, as a feedstock for green H_2 production presents a significant untapped opportunity for the energy sector. By leveraging this readily available resource, the oil and gas industry can potentially unlock a new pathway for the production of green hydrogen, a clean and sustainable energy carrier. This innovative approach can help to reduce the environmental footprint of oil and gas operations, while also contributing to the growth of a low-carbon economy (Odnoletkova and Patzek 2023, Hoecherl et al 2024). However, produced water is often characterized by complex compositions and elevated salinity levels, rendering it unsuitable for direct reuse in traditional agricultural or H_2 production applications. Despite these challenges, leveraging produced water for hydrogen production offers a valuable opportunity to mitigate the strain on freshwater resources and provide a valorization pathway for what would otherwise be considered waste. Furthermore, integrating green hydrogen production with existing oil and gas infrastructure could accelerate the transition to cleaner energy landscapes by repurposing existing assets and expertise. Nevertheless, harnessing produced water for electrolysis introduces unique technological and scientific challenges. The high total dissolved solids (TDS) content, presence of organic compounds, and variability in composition require specialized treatment protocols and robust electrolysis systems capable of handling such feedstocks without compromising efficiency or longevity. Addressing these challenges will be crucial to unlocking the full potential of produced water as a feedstock for green H_2 production and realizing the benefits of such innovative approach (Olaitan et al 2024, Jiang et al. 2023).

Recent advancements in desalination technologies, water treatment methodologies, and electrolyzer designs have begun to address the aforementioned hurdles, paving the way for pilot-scale demonstrations and eventual industrial implementation (Tonget al 2020). The implementation of Zero Liquid Discharge (ZLD) technology (Ayirala et al., 2019; Ayirala et al. 2021) has revolutionized the treatment of produced water in upstream operations (Muhammadet al 2024). By integrating a pretreatment system, the residual oil and dissolved hydrogen sulfide (H_2S) content in hypersaline produced water can be significantly reduced to less than 10 parts per million (ppm) (Alyousef et al. 2024, Ayirala et al. 2024). Subsequently, the pretreated produced water undergoes an evaporation-based "dynamic vapor recovery" process, which enables efficient desalination and yields a high-quality distillate. Notably, the ZLD desalination process achieves a remarkable recovery efficiency of over 75%, resulting in the production of distillate water that consistently meets stringent water quality standards (Muhammad et al 2024). The total dissolved solids (TDS) levels in the treated water are consistently maintained below 200 ppm, demonstrating the efficacy of the distillation process in producing high-quality freshwater (Muhammadet al 2024). The versatility of ZLD distillate water makes it an attractive resource for various upstream applications, including injection water for reservoir re-injection in water flooding, makeup water for improved/enhanced oil recovery (IOR/EOR) processes, and low-salinity water for fracking operations (Ramasamy. et al 2024). Additionally, the treated water can be utilized as wash water for crude oil desalting, as well as a potential water resource for irrigation. Especially, the successful implementation of ZLD technology underscores its potential to provide a reliable and sustainable source of high-quality water for diverse green H_2 production, while minimizing environmental impacts and promoting the conservation of ground water.

In this paper, we investigate the feasibility of utilizing ZLD distillate water, derived from produced water through an advanced ZLD produced water management technology (comprising of a customized pretreatment system and dynamic vapor recovery compression unit), as a potential resource for green hydrogen production via water electrolysis. If the tested ZLD distillate water is proven to be a high-quality feedstock for green hydrogen production, it could unlock a new frontier in sustainability by enabling the effective recycling and reuse of produced water, thereby reducing the waste and promoting a circular economy.

Experimental Set Up

This study investigates the feasibility of utilizing the ZLD distillate water as a sustainable resource for green hydrogen (H_2) production via proton exchange membrane (PEM) and anion exchange membrane (AEM)electrolysis. The distillate ZLD water was generated from produced water through a ZLD treatment process, which significantly reduced the total dissolved solids (TDS) from an initial concentration exceeding 200,000 ppm to less than 200 ppm TDS. The geochemical composition of the resulting ZLD distillate water analysis by ion chromatography (IC) is presented in Table 1, providing some insights into its suitability for H_2 production.

Table 1—Geochemistry of distillate ZLD water

pH	TSS [mg/L]	TDS [mg/L]	CO_3^{2-} [mg/L]	HCO_3^- [mg/L]	Ca^{2+} [mg/L]	Mg^{2+} [mg/L]	K^+ [mg/L]	Na^+ [mg/L]	SO_4^{2-} [mg/L]	Cl^- [mg/L]	TOC [mg/L]
10.4	<5	<196	319	<1.0	<1.0	<1.0	<1.0	<1.0	184	62	229

AEM and PEM cells were assembled using a membrane electrode assembly (MEA) with specific components for testing feasibility of ZLD distillate water and purified ZLD distillate water as green H2 production resource. The following sections describe the preparation of catalyst inks, membrane, and gas diffusion layers, as well as the experimental setup and conditions.

Catalyst Ink Preparation and Coating Process

The catalyst inks for the anode and cathode were formulated by combining catalyst powders, a polymeric binder, and a solvent mixture in predetermined ratios. For the AEM cells, the anode catalyst ink was prepared by dispersing iridium oxide (IrO_2, H2EL-45IrO-S60, Heraeus) in 1:1 volume ratio mixture of ultrapure water and isopropyl alcohol (IPA). The dispersion was stabilized using AP2-HNN8-00-X (Ionomr Innovations Inc.) as the ionomeric binder. The cathode catalyst ink was prepared using an analogous method, with platinum on carbon (Pt/C, Dongwoo Finechem) mixed with the same binder (AP2-HNN8-00-X) in a 1:1 ultrapure water/IPA solution. The prepared inks were applied to gas diffusion layers (GDLs) using a catalyst-coated diffusion layer (CCD) technique to ensure homogeneous catalyst layer deposition. For the PEM cells, the ink formulation was modified by substituting the anion-exchange binder with Nafion D2021 (Chemours), and the catalyst layers were applied directly onto the membrane via the catalyst-coated membrane (CCM) method.

Membrane Preparation and MEA Assembly

The Aemion™ AF3-HWK9-75-X membrane (Ionomr Innovations Inc.), with a nominal thickness of 80 μm, was positioned between the anode and cathode catalyst layers to form the MEA. Prior to assembly, the membrane underwent a pre-treatment process consisting of immersion in 1 M KCl solution (Sigma-Aldrich) for 5 hours, followed by soaking in 1 M KOH solution (Samchun Chemicals) for a minimum of 12 hours. This dual-step conditioning was performed to enhance the membrane's ionic conductivity and chemical stability. For PEM electrolysis cells, a GORE™ PEM M275.80 membrane (Gore & Associates) was employed in place of the AEM.

Gas Diffusion Layer Integration

Gas diffusion layers (GDLs) were applied to both sides of membrane electrode assembly (MEA) to facilitate effective gas transport. The anode side utilized a nickel felt diffusion layer (Currento® PTL INC-79/600, Bekaert), while the cathode side employed a molten graphite paper (JNP 20, JNTG) with a thickness of 0.41 mm. These materials were chosen based on their chemical compatibility with the respective electrode environments and their capacity to support efficient gas diffusion and water management. Prior to assembly, the nickel felt was treated by immersion in 1 M HCl solution for 20 minutes to remove surface oxides and enhance surface activity, thereby improving electrochemical performance. For PEM electrolysis, titanium fiber paper (Currento®PTL Ti-56/250, Bekaert) and carbon paper (GDS22100, AvCarb) were used as anode and cathode diffusion layers, respectively, due to their stability under acidic conditions and suitability for high-performance PEM operation.

Electrolyte Solution and Temperature Control

1 M KOH solution prepared with deionized (DI) water or raw/treated Zero Liquid Discharge (ZLD) distillate water was used as the electrolyte, providing the necessary ionic conductivity for cell operation. The electrolyte temperature was maintained at 70°C for AEM cells and 60°C for PEM cells to optimize ionic conductivity and electrode kinetics. The entire cell was operated at a stable temperature of 70°C and 60°C for AEM and PEM cells, respectively, to ensure consistent electrochemical performance.

Flow Rate and Electrolyte Circulation

The electrolyte flow rate was set to 10 mL/min, and the electrolyte was circulated on both the anode and cathode sides to maintain uniform distribution and remove any generated gases or reaction byproducts.

Current Density and Current Voltage (I-V) Measurement

The current density was varied between 0.1 and 2 A/cm^2 for AEM cells and 0.1 and 3 A/cm^2 for PEM cells to generate an I-V curve, which was used to evaluate the cell's electrochemical performance under different load conditions.

Durability Test Conditions

The cell's durability was assessed using 1 M KOH electrolyte solution with purified Korean water or raw/treated ZLD water. The durability test was conducted at a flow rate of 10 mL/min, applying a constant current density of 0.5 A/cm^2. For PEM cells, the raw/purified ZLD water was used without any additions.

The process flow diagram and configuration of experimental setup used for I-V and durability measurements are shown in Figure 1 and 2. The setup consists of a glass oxygen separator (10 L capacity), a circulation pump (Cole-Parmer MasterFlex L/S), and an electrolyzer cell. ZLD distillate or purified ZLD distillate water is introduced into the oxygen separator, where it is heated to 60°C. The heated water is then circulated to the electrolyzer cell via the circulation pump. Unreacted water is continuously recirculated back to the oxygen separator, where dissolved oxygen is vented to the atmosphere. Hydrogen produced during electrolysis is vented continuously, as shown in Figure 1(a). For systems utilizing purified ZLD distillate water, an additional process step is included, in which raw water is first treated through a deionized (DI) water generator prior to being fed to the oxygen separator, as shown in Figure 1(b). Figure 2 shows the configuration of I-V measurement and durability test.

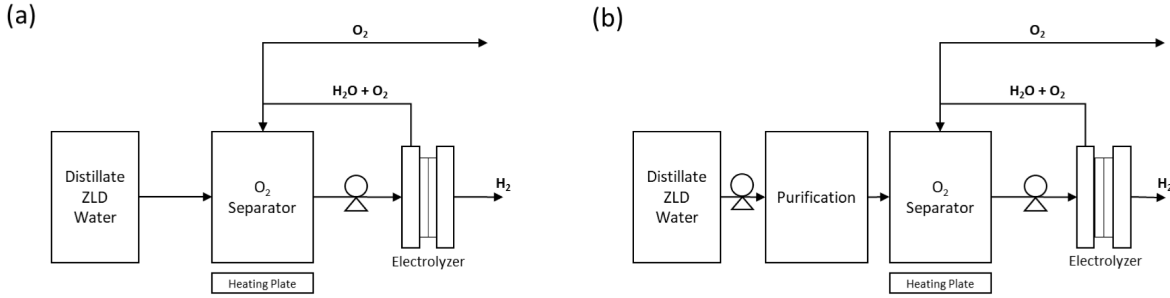

Figure 1—Process flow diagram of water performance evaluation. (a) is for ZLD distillate water and (b) is for purified ZLD distillate water.

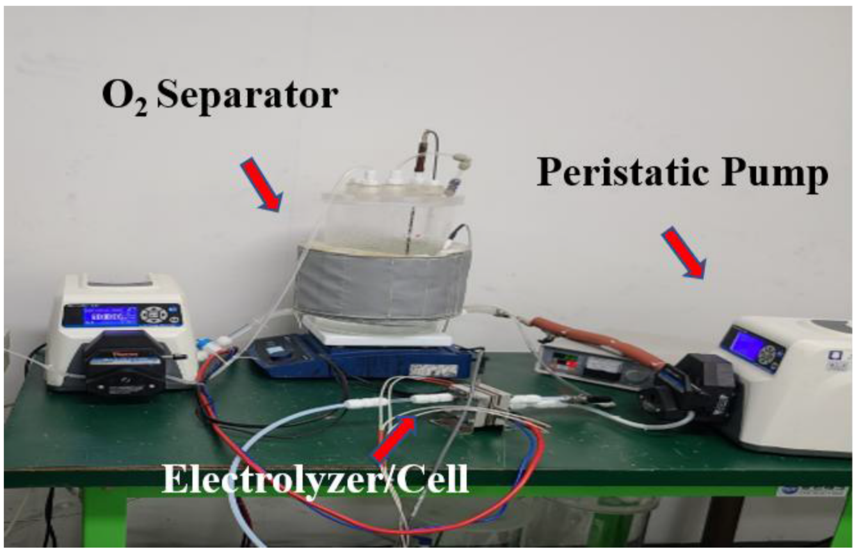

Figure 2—Experimental set-up for I-V measurement and durability test

Results and Discussion

Purified ZLD distillate water vs, ZLD distillate water

As a reference, the electrochemical properties of purified ZLD distillate water were investigated using an AEM cell. The I-V curve, measured at 60°C using 1 M KOH electrolyte, is presented in Figure 3. The curve exhibits a positive linear relationship between current density and voltage, indicating that the purified distillate water enables efficient operation of the AEM cell over a wide range of current densities (0.1-2 A/cm^2). The voltage requirements at 1 A/cm^2 and 1.5 A/cm^2 were found to be 1.80 V and 1.94 V, respectively. The steady increase in voltage with increasing current density is attributed to the combined effects of ohmic resistance and concentration losses, which are typical in electrochemical cells under load. The linearity of I-V curve suggests that the voltage is influenced by a combination of activation and ohmic overpotentials, characteristic of electrochemical cells operating under increasing load conditions. These results serve as baseline for evaluating the electrochemical properties of ZLD distillate water, which will be discussed in the following section. The comparison between the purified distillate water and the raw distillate ZLD water will enable the assessment on the suitability of the latter as a feedstock for green H_2 production.

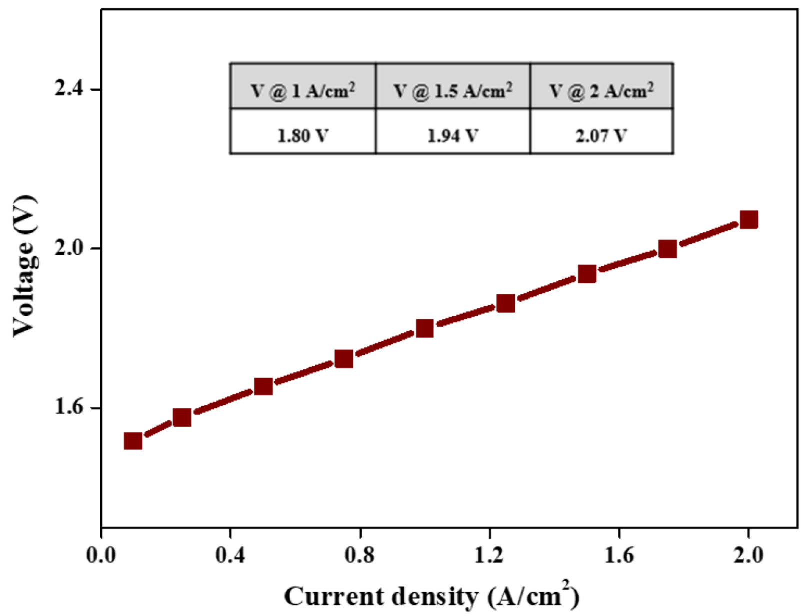

Figure 3—I-V curve for purified ZLD distillate water using AEM cell at current densities ranging from 0.1 to 2.0 A/cm²

The quality of ZLD distillate water was assessed through a comprehensive electrochemical evaluation using AEM cell. The I-V curve, measured at a controlled temperature of 60°C, provided valuable insights into the electrochemical properties of the distillate water. As illustrated in Figure 4, the I-V curve reveals a positive linear correlation between current density (A/cm^2) and voltage (V) for the AEM cell using 1 M KOH electrolyte with ZLD distillate water. The results demonstrate that ZLD distillate water exhibits a stable and consistent electrochemical behavior, with a linear relationship between current density and voltage. This linearity suggests that the voltage is influenced by a combination of activation and ohmic overpotentials, characteristic of electrochemical cells operating under increasing load conditions. Notably, the ZLD distillate water supports a wide range of current densities, from 0.1 to 2 A/cm^2, with relatively low voltage requirements. Specifically, the voltage requirements at current densities of 1 A/cm^2 and 1.5 A/cm^2 were found to be 1.84 V and 1.97 V, respectively. The relatively high voltage values at increased current densities could suggest resistance within the cell or limitations in the conductivity of the electrolyte or the membrane. These findings indicate that ZLD distillate water is of high quality, making it an ideal feedstock for green hydrogen (H_2) production. The electrochemical evaluation provides a comprehensive understanding of ZLD distillate water's properties and behavior, which is essential for optimizing its performance in green H_2 production process. The results of this study thereby demonstrate the potential of ZLD distillate as a high-quality water resource for green H_2 production, highlighting its suitability for use in AEMWE.

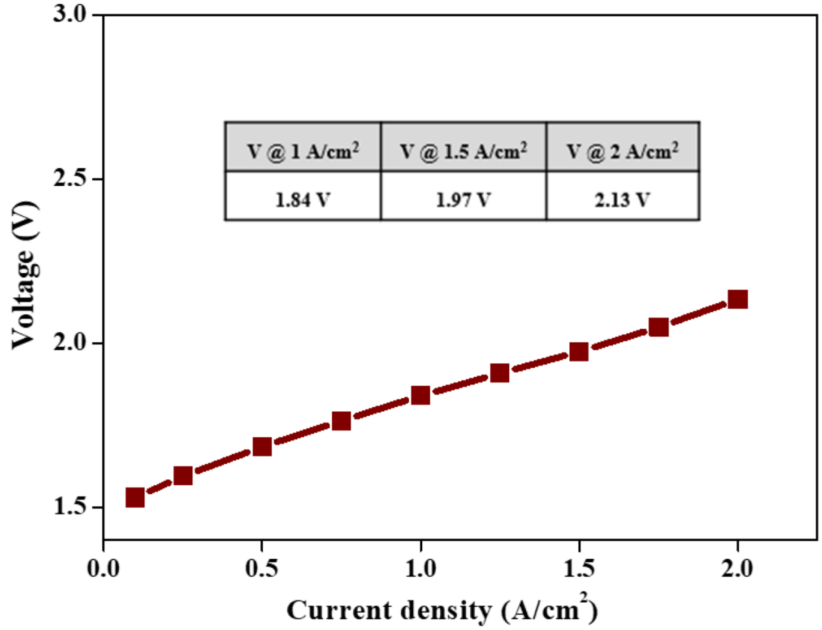

Figure 4—I-V curve for raw distillate ZLD water using AEM cell at current densities ranging from 0.1 to 2.0 A/cm²

The durability test was also conducted for evaluating raw ZLD distillate water. Figure 5 shows the results obtained from durability test for an AEM cell operated with raw ZLD distillate water mixed with 1 M KOH as the electrolyte. When the current density is set at 0.5 A/cm², the initial voltage is approximately 1.67 V. This indicates the starting potential needed to maintain the reaction at this current density. Over 144 h, there is a gradual increase in voltage, reaching approximately 2.12 V at the end of the test (t = 144 h). Such trend suggests that the cell requires a higher voltage to sustain the same current density. The phenomenon is generally attributed to performance degradation or increased resistance in the cell. The degradation rate is reported as 3,125 µV/h, calculated from the change in voltage over time. This value quantifies the rate at which the cell performance declines, likely due to factors such as membrane fouling, electrolyte impurities, or electrode degradation. The optimized water treatment and purification strategies need to be adopted to mitigate these issues and ensure the longevity/efficiency of the cell.

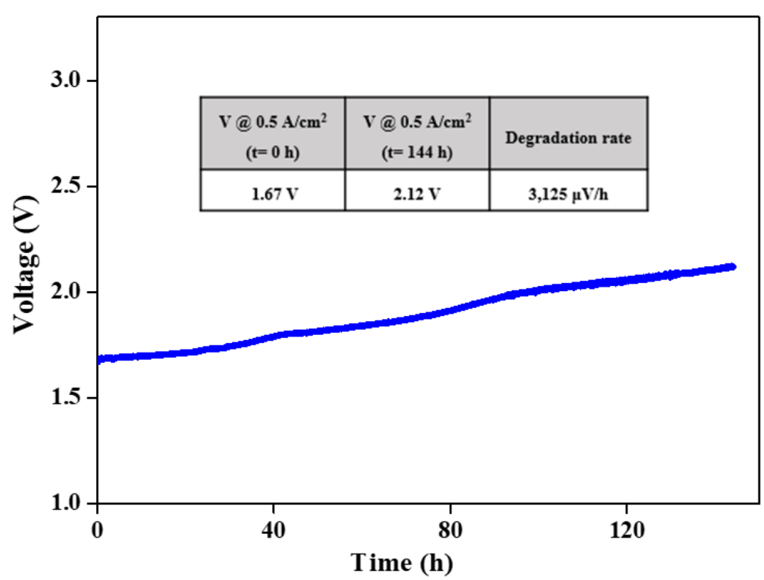

Figure 5—Durability test results for raw ZLD distillate water using AEM cell at a current density of 0.5 A/cm² for 144 h.

AEM vs. PEM. In addition, the electrochemical properties of raw ZLD distillate water were investigated using PEM cell with the objective of comparing its performance with that of AEM cell. As shown in Figure 6, the voltage increased sharply from 1.53 to 2.11 V as the current density rose from 0.1 to 0.4 mA/cm². Thereafter, the voltage increased at a slower rate with further increments in current density. This indicates that the hydrogen generation from raw ZLD distillate water using PEM cells is not suitable because it is only limited to low current densities. AEM is often used for desalination and water treatment applications, and it can tolerate a wide range of feed water qualities, including high levels of total dissolved solids (TDS), turbidity, and organic matter (Tong et al 2020, Lindquist2020). PEM technology, on the other hand, is commonly used for water electrolysis and fuel cell applications, and it typically requires high-purity feed water with low levels of contaminants.

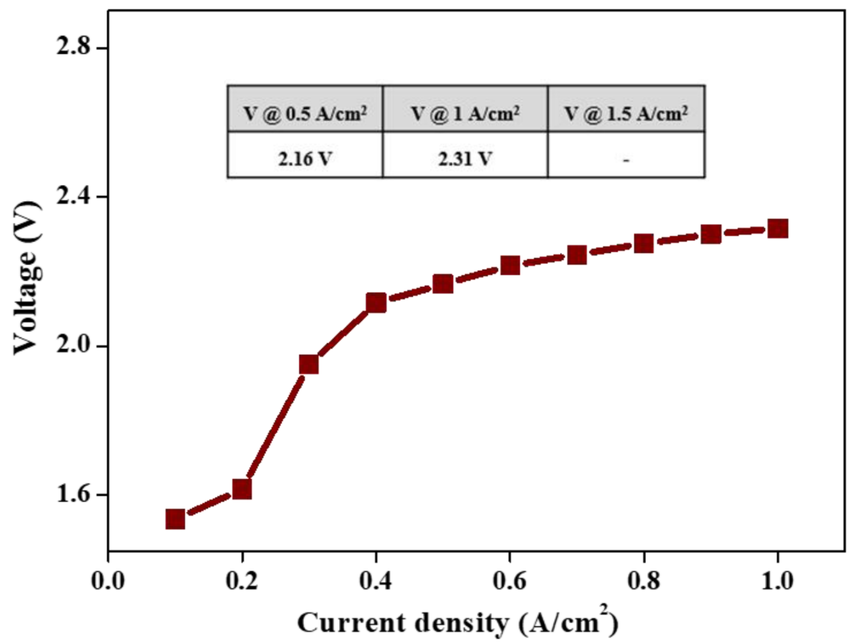

Figure 6—I-V curve for ZLD distillate water using PEM cell at current densities ranging from 0.1 to 2.5 A/cm².

Conclusions

This study investigates the feasibility of utilizing treated produced water as a feedstock for green hydrogen production. The performance of AEMWE and PEMWE were evaluated using raw ZLD distillate water. The key findings are summarized as follows:

- AEMWE exhibited compatible performance for hydrogen production when operated with raw ZLD distillate water, demonstrating its potential as a high-quality feedstock.
- The durability test results revealed a decline in performance, characterized by increased resistance within the cell, which suggests the occurrence of membrane fouling, electrolyte impurities, or electrode degradation. To mitigate these issues, it is suggested to implement optimized water treatment and purification strategies, ensuring the longevity and efficiency of the cell.
- While hydrogen production from ZLD distillate water using PEMWE is feasible performance, the results show that ZLD distillate water only enables limited performance at low current densities due to the higher purity requirements of PEMWE compared to AEMWE.
- The study confirms that hydrogen production from ZLD distillate water can be achieved using AEMWE. Furthermore, precise control of impurity levels in the ZLD distillate water is expected to further enhance the performance of AEMWE, highlighting the need for optimized water treatment and purification strategies.

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